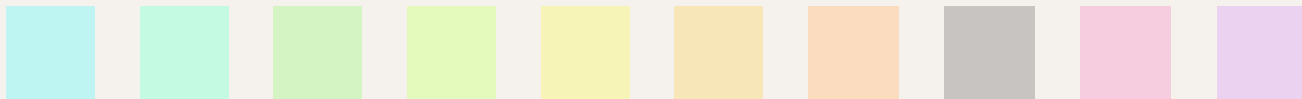


Introduction to Web Development

מבוא לבניית מערכות ממוחשבות מבוססות אינטרנט – הרצאה 1 – מבוא



About the Course

Lecturer

Itzik
Gurevich

TA

Paz Arad

Office Hours

tentative

Prerequisites

DB
OOP
SAD

Syllabus

Project

3 parts
40%

Course Objectives

- Understanding basic architecture and infrastructure of the internet
- Getting familiar with basic web development technologies
- Experiencing with full web development project



Full Stack Web Development

Client Side



Introduction & Git

HTML, HTML 5

CSS

Bootstrap

JavaScript

DOM

Client-Server

Node.js

Express

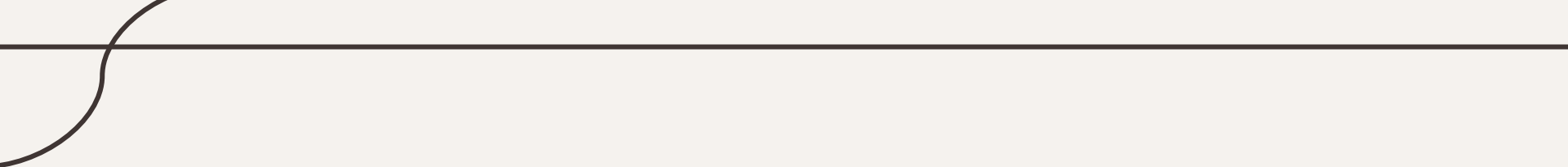
DB (MySQL)

Server Side



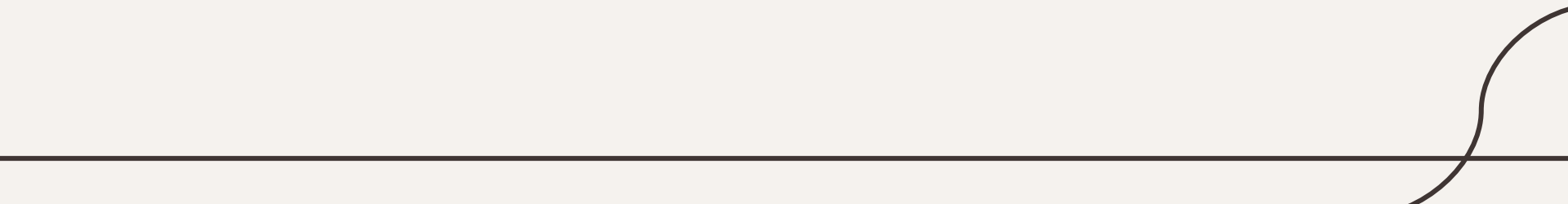
express





01

Internet- A Brief History



The Internet in 3 Key Moments

- **The Big Idea (1950s-70s):**
 - Sputnik is launched (1957)
 - Advanced Research Projects Agency (ARPA) is created (1958)
 - Packet switching- Leonard Kleinrock (1961)
 - Intergalactic network- J.C.R Licklider (1962)
 - The mother of all demos- Douglas Engelbart (1968)
 - @ (user at domain)- Ray Tomlinson (1972)
 - Top level domains- .gov, .org, .com...- Elizabeth Feinler (1974)

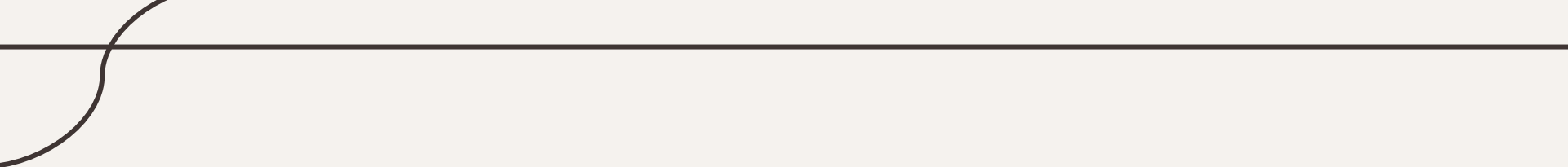
STATEMENT ONLY. WORD WORD WORD WORD WORD WORD
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1



The Internet in 3 Key Moments

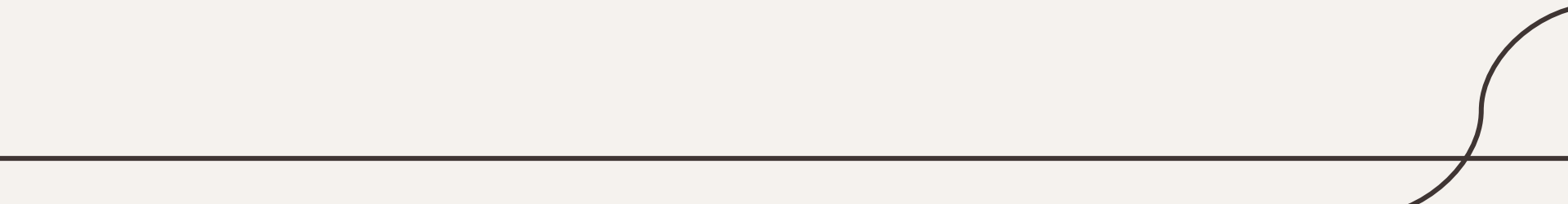
- **The Breakthrough (1990s):**
 - Tim Berners-Lee invents the World Wide Web, making the internet accessible.
- **The Explosion (2000s+):**
 - Google, social media, and smartphones bring the internet into daily life.

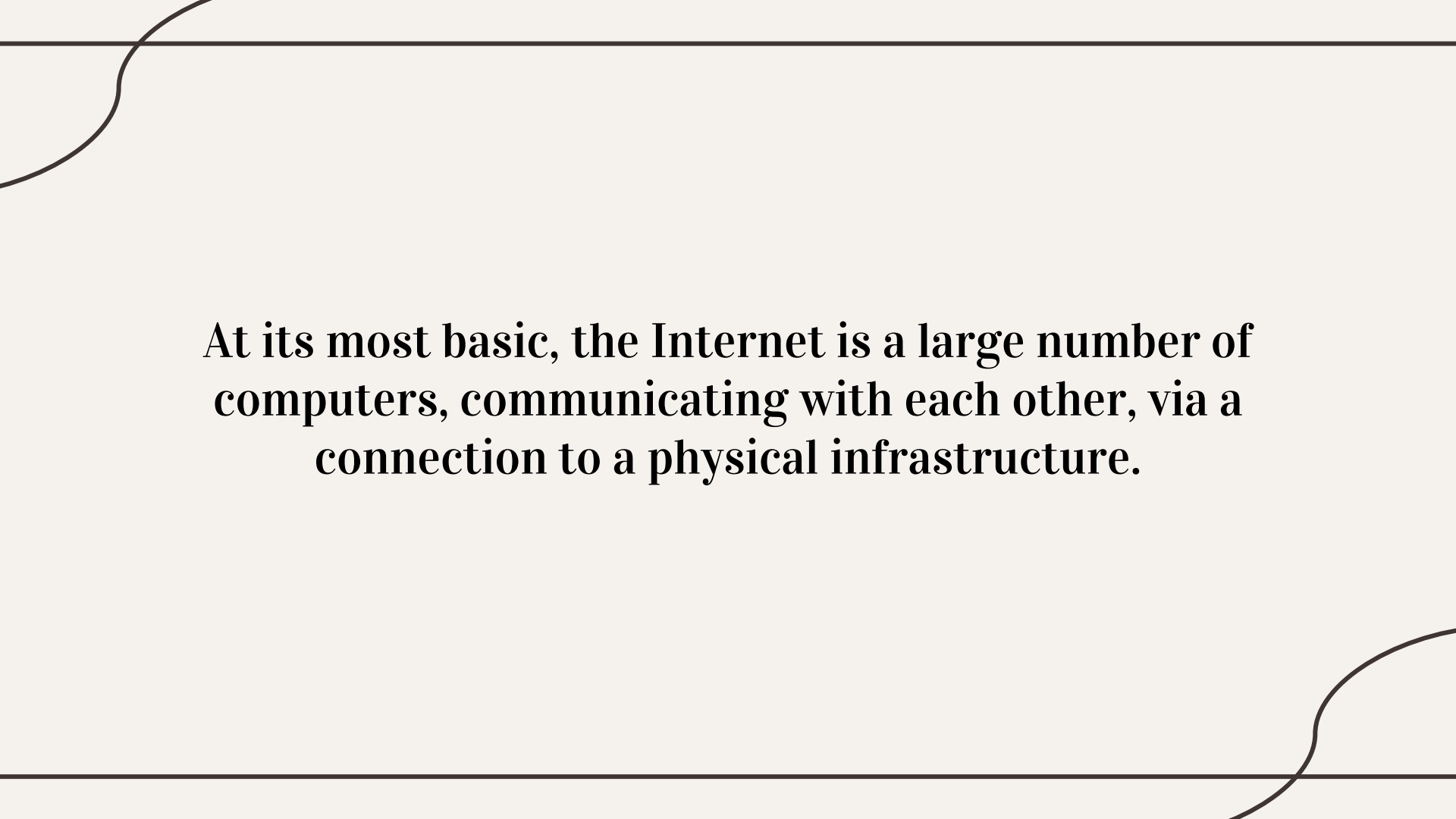


02

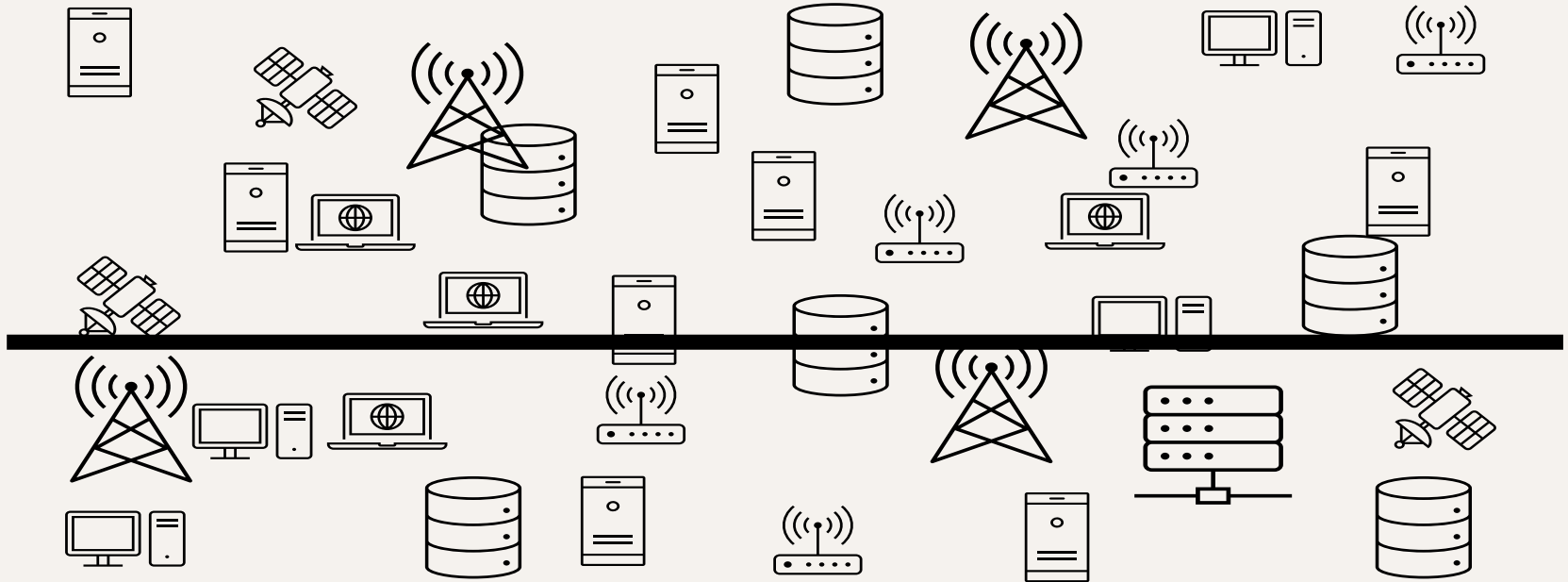
How Does the Internet Work?

Components and concepts

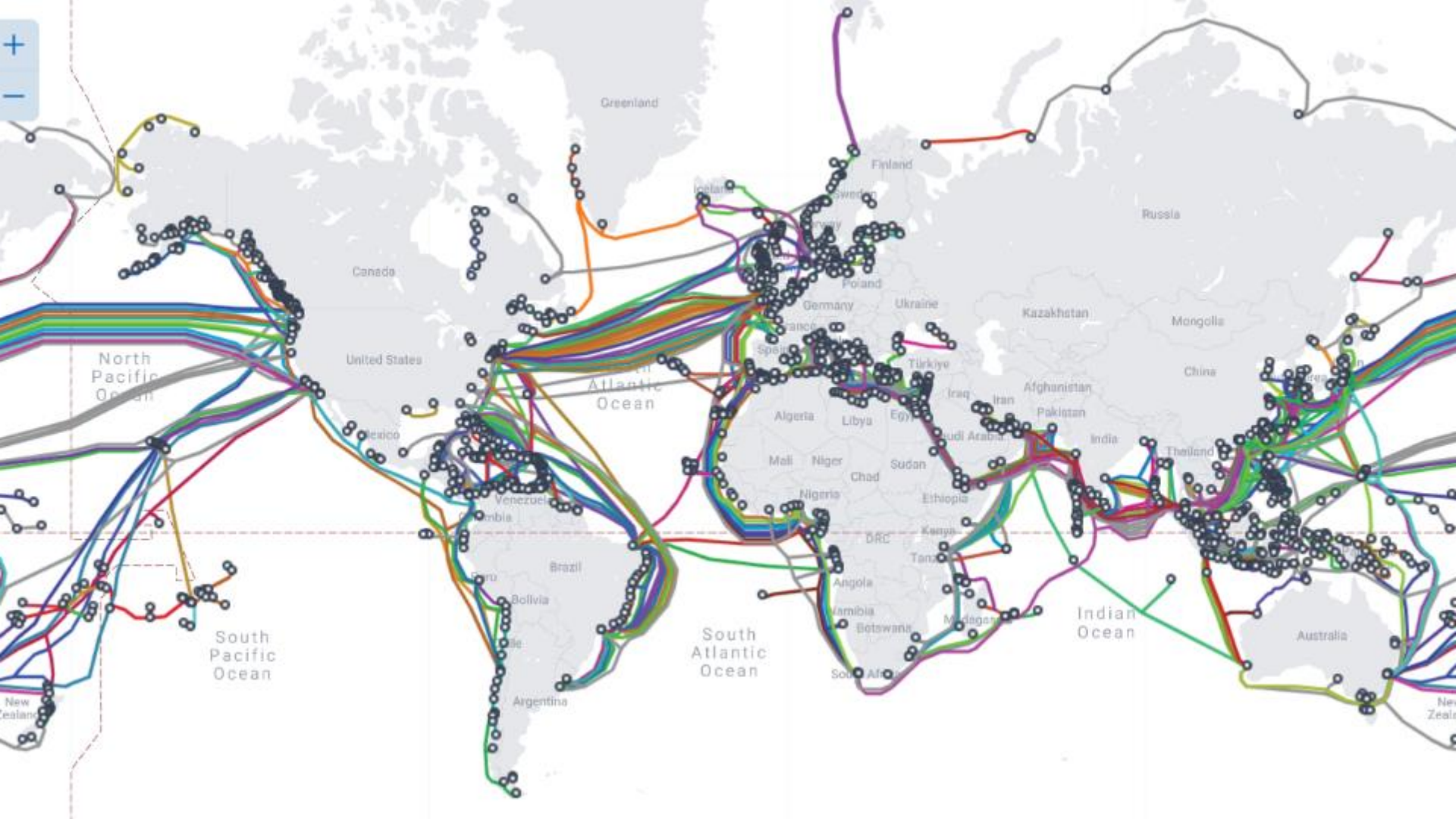


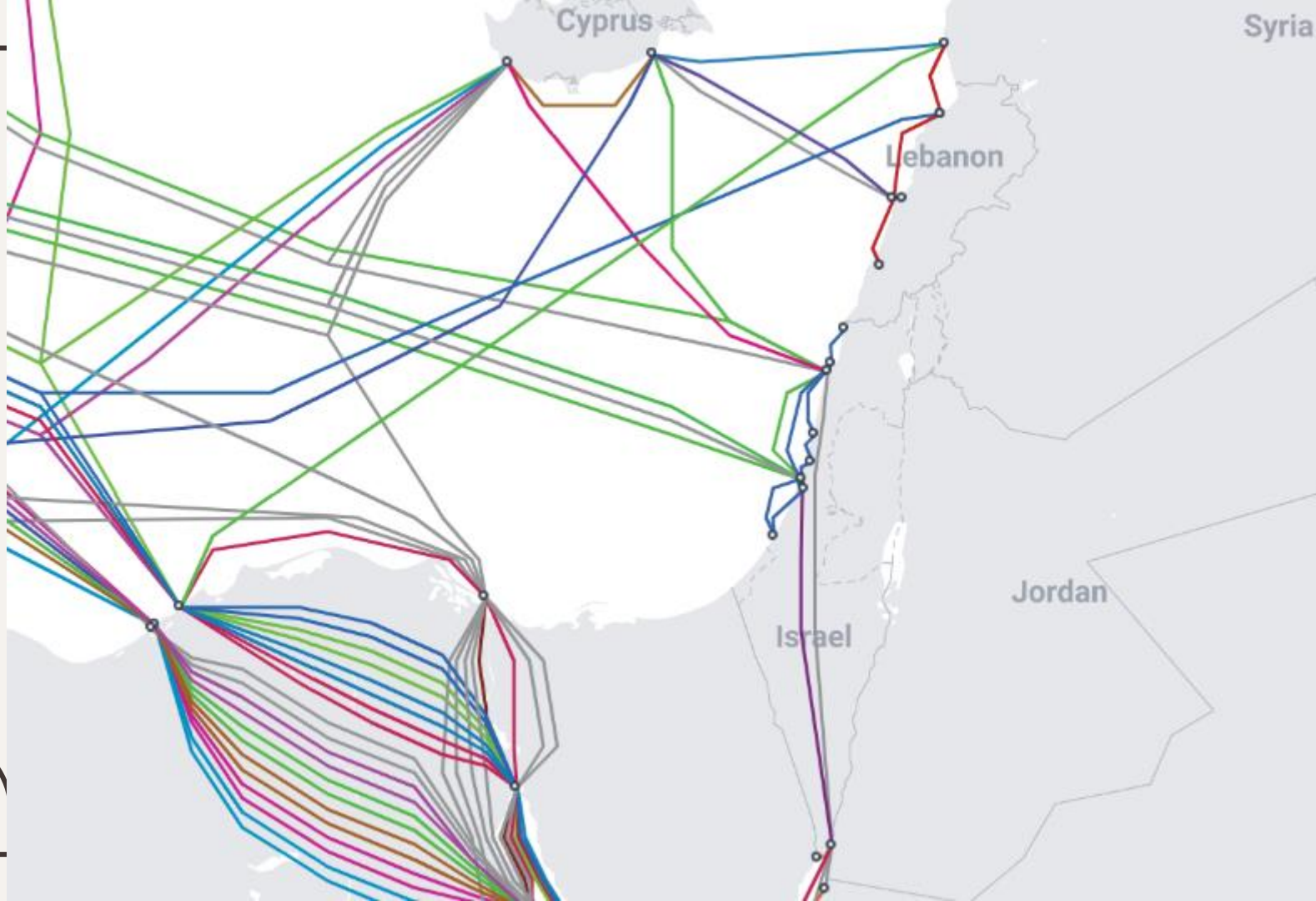


At its most basic, the Internet is a large number of computers, communicating with each other, via a connection to a physical infrastructure.



The internet is a wire and everything that is connected to it

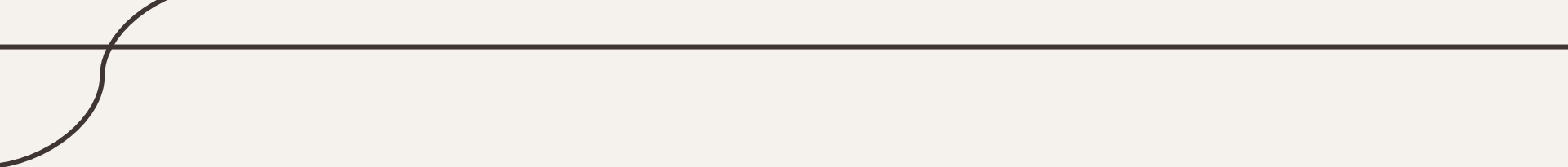






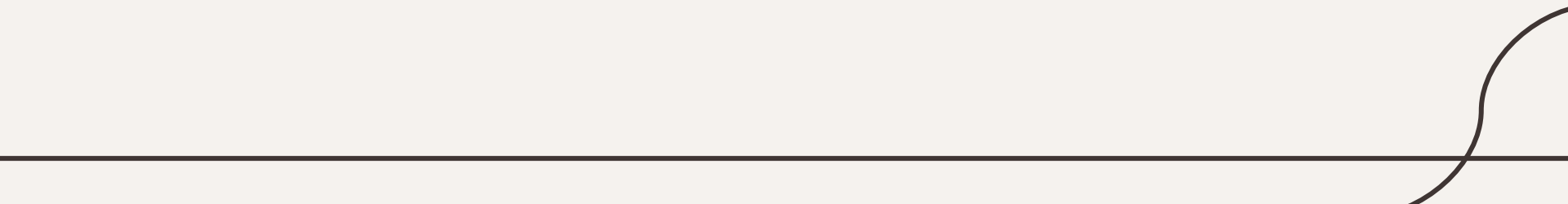
The “Dunant” Cable owned by Google, Connects US and France





03

Protocols, URL



IP- Internet Protocol

What is a Protocol?

- A set of rules that define how data is transmitted and received over a network.

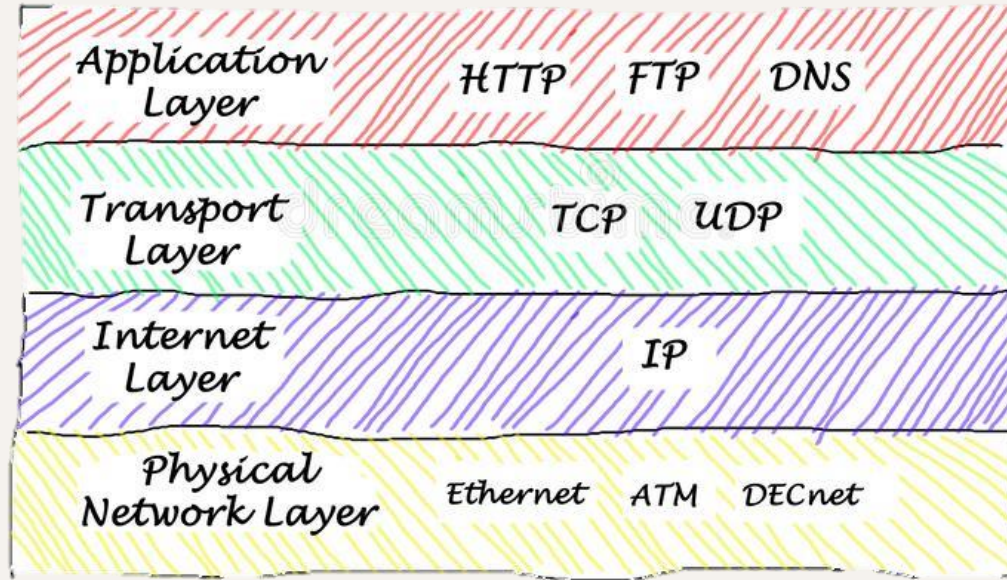
What is Internet Protocol (IP)?

- The fundamental system of the internet, ensuring that data packets reach the correct destination.
- Works by assigning unique addresses to devices and routing data efficiently.

Examples of Protocols Using IP:

- HTTP & HTTPS - Web browsing
- SMTP & IMAP – Emails
- FTP - File transfers

Internet Protocol



HTTP- Hyper Text Transfer Protocol

What is HTTP?

- A communication protocol used for transferring web pages and data over the internet.
- Defines how requests and responses are formatted and transmitted.

How HTTP Works:

1. The client (browser) **requests** a resource (webpage, image, etc.).
2. The server receives the request and processes it.
3. The server sends a **response** (HTML, CSS, JS, etc.) back to the client.
4. The client renders the webpage using the received data.

HTML- Content

CSS- Design

JavaScript- Functionality

HTML



CSS



JS



HTTP- Hyper Text Transfer Protocol (2)

Key Features:

- **Stateless:** Each request is independent- no memory of previous interactions.
- **Methods:**
 - GET → Retrieve data (e.g., web pages)
 - POST → Send data (e.g., form submissions)
 - PUT → Update a resource
 - DELETE → Remove a resource

URL- Uniform Resource Locator

web address pointing to a specific website, a web page, or a document on the internet.

<protocol>://<hostname>/<relative-path>?<query-parameters>

URL- Uniform Resource Locator

<**protocol**>://<hostname>/<relative-path>?<query-parameters>

protocol

For instance: http, https. mailto, etc.

URL- Uniform Resource Locator

<protocol>://<**hostname**>/<relative-path>?<query-parameters>

Hostname

For instance: www.google.com

- Subdomain (www - world wide web)
- Domain (google, bgu, etc.)
- TLD – Top Level Domain (.com, co.il, etc.)

URL- Uniform Resource Locator

<protocol>://<hostname>/<**relative-path**>?<query-parameters>

Relative path

- Represents a specific path on the specific server
- A relative path refers to a location that is relative to a current directory

URL- Uniform Resource Locator

<protocol>://<hostname>/<relative-path>?<**query-parameters**>

query-parameters

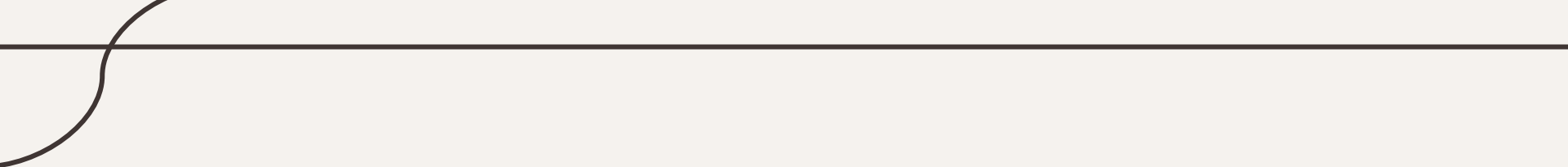
?key1=value1&key2=value2&...&keyN=valueN

- Translating
- Searching
- Filtering
- Etc.

URL- Uniform Resource Locator

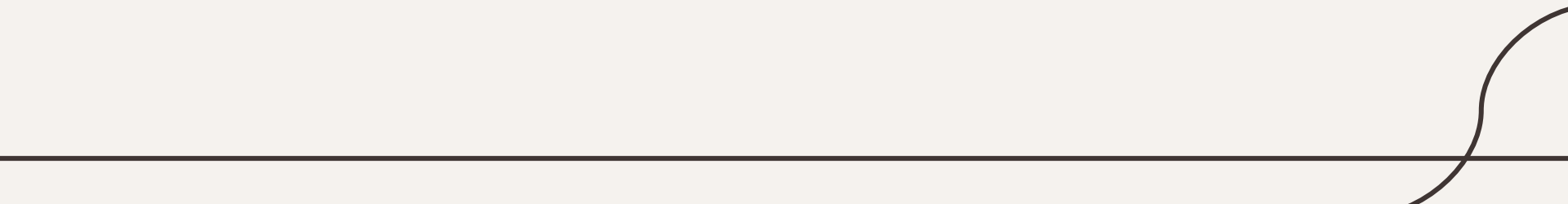
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<https://www.google.com/search?...&q=dua+lipa&oq=dua+lipa&...>



04

Technical Set Up in the Course



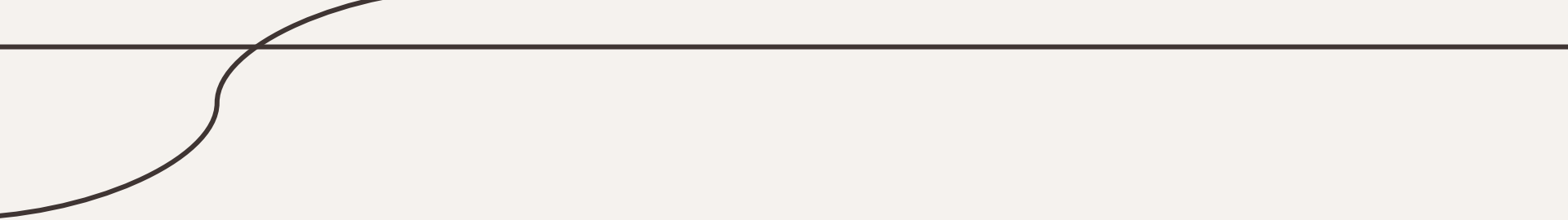
Technical Set Up

1. Install Code editor
(Visual Studio code).
2. Install Git
3. Sign up to GitHub

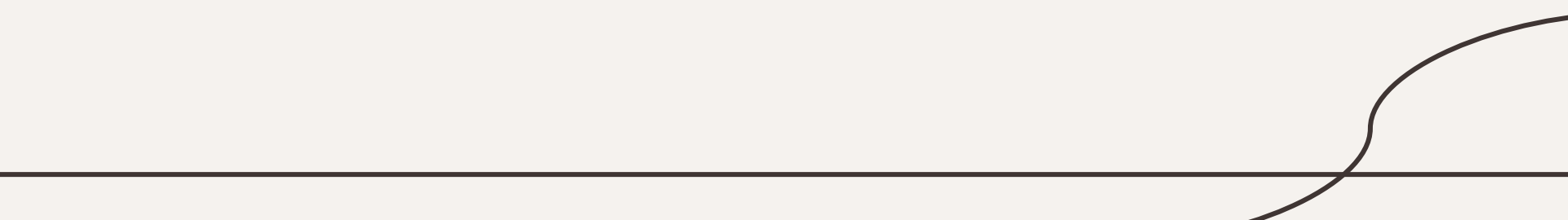


Visual Studio





Introduction to Version Control, Git and GitHub



Introduction to Version Control

- What is version control?



What is Git?

- A distributed version control system
- Git helps developers collaborate efficiently
- Local vs. remote repositories

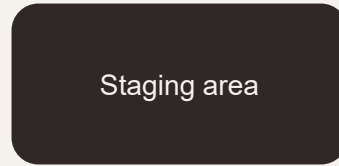
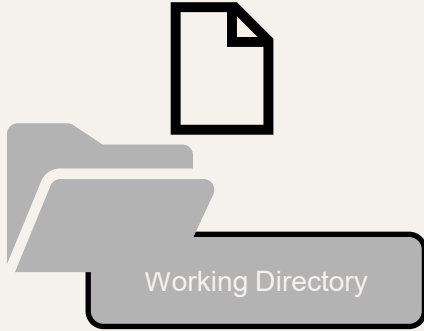
Setting Up Git

- Installing Git (Windows, Mac, Linux)
- Configuring Git:

```
git config --global user.name "Your Name"  
git config --global user.email "your.email@example.com"
```

Essential Git Commands

`git init`



`git add "file name"`

`git commit`

`git checkout`



Essential Git Commands

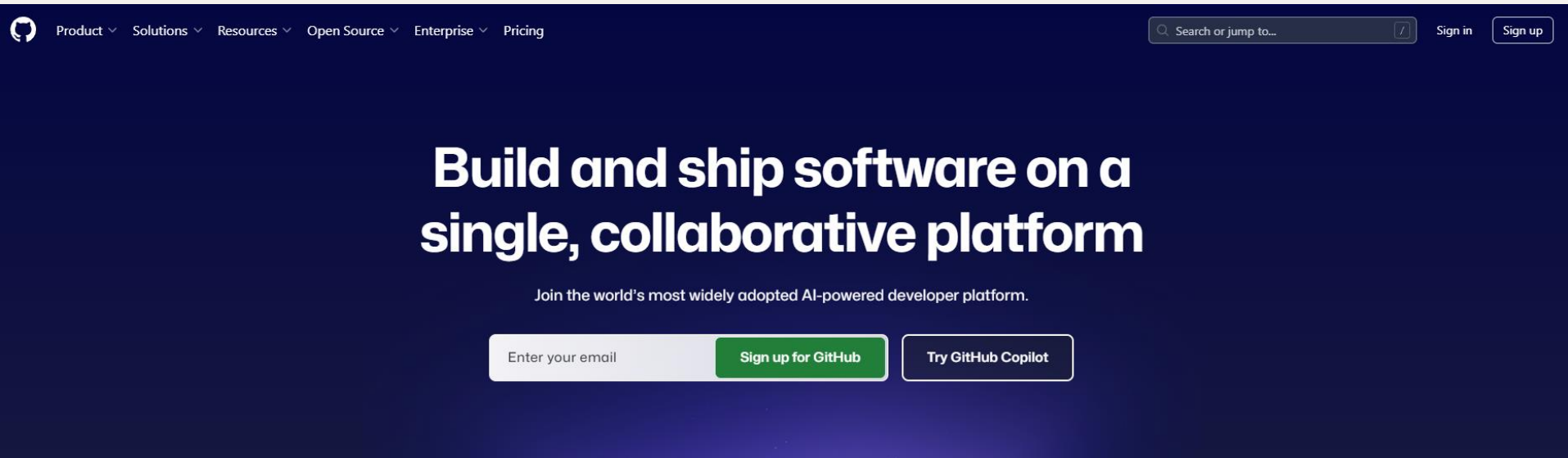
- Initializing a repository: `git init`
- Checking status: `git status`
- Adding files to staging: `git add filename`
- Committing changes: `git commit -m "commit message"`
- Viewing commit history: `git log`
- Roll back to the last position committed:
`git checkout <commit hash> -- "file name"`
- Check differences between versions (before committing):
`git diff "file name"`

What is GitHub?

- GitHub as a cloud-based Git repository hosting service
- Public vs. private repositories

Opening an account in GitHub

- Browse into github.com

A screenshot of the GitHub homepage. The header is dark blue with the GitHub logo on the left, navigation links (Product, Solutions, Resources, Open Source, Enterprise, Pricing) in the center, and a search bar, 'Sign in' link, and 'Sign up' button on the right. The main content area has a dark blue background with white text. The headline reads 'Build and ship software on a single, collaborative platform'. Below it is a sub-headline 'Join the world's most widely adopted AI-powered developer platform.' and three buttons: 'Enter your email' (light gray), 'Sign up for GitHub' (green), and 'Try GitHub Copilot' (white with a dark border).

Product ▾ Solutions ▾ Resources ▾ Open Source ▾ Enterprise ▾ Pricing

Search or jump to... [1] Sign in Sign up

Build and ship software on a single, collaborative platform

Join the world's most widely adopted AI-powered developer platform.

Enter your email Sign up for GitHub Try GitHub Copilot

Essential Git Commands



```
git push -u <remote name> <branch name>
```

Creating and Managing a Repository on GitHub

- Creating a repository
- Cloning a repository: `git clone repository_url`
- Connecting a local repository to GitHub:

```
git remote add origin "remote_repository_url"
```

- Pushing changes:

```
git push -u origin master\main
```

Working with Branches

- What are branches in Git?
- Creating a branch: `git branch branch_name`
- Switching to a branch: `git checkout branch_name`
- Merging branches: `git merge branch_name`

Best Practices for Using Git and GitHub

- Commit frequently with clear messages
- Use branches for different features
- Keep repositories organized

05

Plan and Design

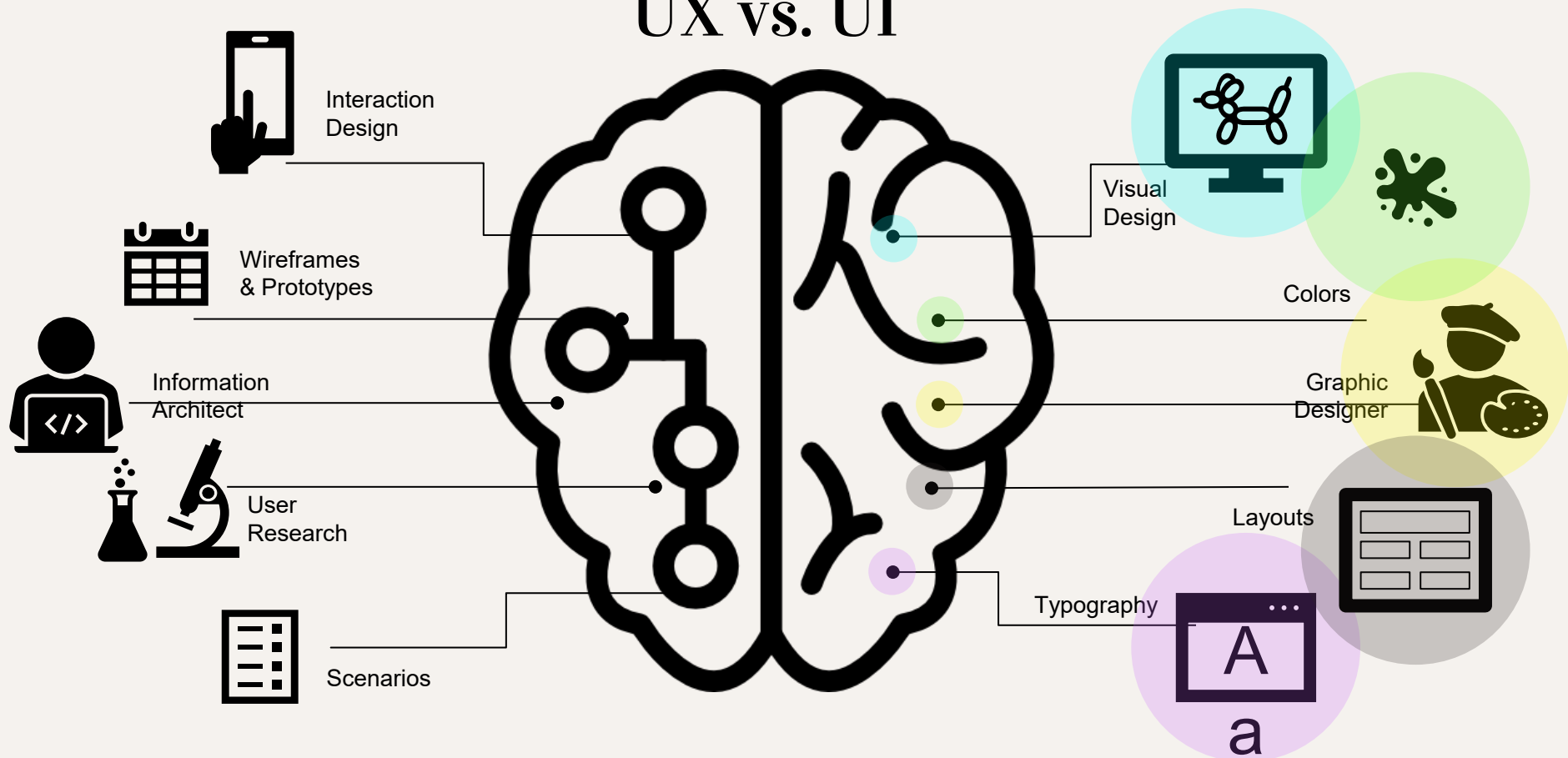
UX/UI

Requirements

Specifications

Documentation

UX vs. UI



UX Requirements

1. who is the user?
2. What need of the user the app is going to satisfy?
3. What is the service the app provides to the user?
4. What is the exact process (procedure) of that service?
What is the content or data needed?
5. What is the database structure needed?

UI Specification

- A UI specification defines the rules of engagement for a user interacting with a specific page on a website or screen within an application
- Visual overview of the screen
 - Break the screen up into sections
 - Within each section, look for the **display rules**
- Consider **messaging**
- Evaluate the **links**

Pages:

- Home Page (/)
 - Header (logo, menu -> about, contact, [profile <- if logged in], login/logout)
 - Simple hero shot and marketing message
 - Super obvious call to action above the fold to enter your email address to get started
 - Simple thank you message that appears in place of the field after a valid email is entered.
 - "About us" content below the fold
 - Intro content (title/couple sentences)
 - 4 sections (either two by two - or 4 columns) - Meetings, Agenda Items, Action Items, Users - each with their own visual icon - each allowing a paragraph of text
 - Social Media Widgets (FB facepile & Latest Tweets)
 - Footer
- Contact Us (/contact)
 - Address / Email / Social Media / Contact Form
- Login (/login)
 - Simple, centered login form - email & password
- Finalize Signup (/finalize_signup)
 - Simple, centered form like the login page, but for First, Last, Password, Password Confirmation

Documentation

- Documenting every stage of your work might seem tedious, nevertheless it is extremely important for the following reasons:
 1. development
 2. maintenance
 3. knowledge transfer